

Marketing Mix Model: Error Metrics & Channel Parameters

Model Identification & Error Metrics

- **solID**: Solution identifier for this specific model run
- **mape**: Mean Absolute Percentage Error - average percentage difference between predictions and actuals
- **decomp.rssd**: Decomposition residual sum of squared deviations
- **nrmse**: Normalized Root Mean Square Error - overall model prediction error
- **nrmse_test**: NRMSE for the test dataset - measures out-of-sample performance
- **nrmse_train**: NRMSE for the training dataset - measures in-sample fit
- **nrmse_val**: NRMSE for the validation dataset - measures model generalization

Adstock Parameters (Theta θ)

Controls the decay rate of marketing effects over time

Channel-specific decay rates:

- **ADNW1_thetas, ADNW2_thetas, ADNW3_thetas**: Ad network decay parameters
- **AFF_thetas**: Affiliate marketing decay parameter
- **BRAND_thetas**: Brand marketing decay parameter
- **DOUGA_thetas**: Video advertising decay parameter
- **GAMEPR230324_thetas, GAMEPR230727_thetas**: Game promotion decay parameters
- **INFLU230402_thetas, INFLU230404_thetas, INFLU230408_thetas, INFLU230821_thetas**: Influencer campaign decay parameters
- **SEM_thetas**: Search engine marketing decay parameter
- **SNS_thetas**: Social network marketing decay parameter

Saturation Parameters (Alpha α)

Controls the diminishing returns curve shape

Channel-specific saturation rates:

- **ADNW1_alphas, ADNW2_alphas, ADNW3_alphas:** Ad network saturation parameters
- **AFF_alphas:** Affiliate marketing saturation parameter
- **BRAND_alphas:** Brand marketing saturation parameter
- **DOUGA_alphas:** Video advertising saturation parameter
- **GAMEPR230324_alphas, GAMEPR230727_alphas:** Game promotion saturation parameters
- **INFLU230402_alphas, INFLU230404_alphas, INFLU230408_alphas, INFLU230821_alphas:** Influencer campaign saturation parameters
- **SEM_alphas:** Search engine marketing saturation parameter
- **SNS_alphas:** Social network marketing saturation parameter

Lagged Effect Parameters (Gamma γ)

Controls the delay before marketing effects begin

Channel-specific lag parameters:

- **ADNW1_gammas, ADNW2_gammas, ADNW3_gammas:** Ad network lag parameters
- **AFF_gammas:** Affiliate marketing lag parameter
- **BRAND_gammas:** Brand marketing lag parameter
- **DOUGA_gammas:** Video advertising lag parameter
- **GAMEPR230324_gammas, GAMEPR230727_gammas:** Game promotion lag parameters
- **INFLU230402_gammas, INFLU230404_gammas, INFLU230408_gammas, INFLU230821_gammas:** Influencer campaign lag parameters
- **SEM_gammas:** Search engine marketing lag parameter
- **SNS_gammas:** Social network marketing lag parameter

Model Classification

- **cluster:** Group classification for this model solution
- **n:** Number of observations used
- **top_sol:** Indicator if this is identified as a top-performing solution

Summary of Parameter Importance

The document breaks down:

1. **Model identification and error metrics** - These show how well your model is performing overall and across different data subsets
2. **Adstock parameters (Theta θ)** - These control how long marketing effects last after the initial exposure, with higher values meaning longer-lasting effects
3. **Saturation parameters (Alpha α)** - These determine how quickly diminishing returns set in as you increase spending on a channel
4. **Lagged effect parameters (Gamma γ)** - These capture any delay between marketing activity and its impact on sales
5. **Model classification metrics** - These identify the model solution and its quality

These parameters are crucial for understanding the dynamics of each marketing channel in your model - how quickly they work, how long their effects last, and at what point additional spending yields diminishing returns.